TEXORA · SESSIONAL NOTES · STUDY SHEET

Industrial Chain Stitch Machines and Stitch Samples

How the two-thread chain stitch (type 401) is formed with a looper instead of a bobbin, how chain-stitch samples are produced and judged, and the full comparison with the lock stitch of Study 7 that decides which machine a seam belongs on.

TE-2208 · APPAREL MANUFACTURING TECHNOLOGY-1 SESSIONAL · STUDY 8 · LEVEL 2 TERM 2

1. Aim and place of the study

Aim: to study the formation of the chain stitch on an industrial chain stitch machine, produce stitch samples, and compare the chain stitch with the lock stitch in formation, appearance, security and use. The comparison is the working knowledge: a technologist meets both machines on every sewing floor and must know at sight which stitch a seam carries and why.

2. The chain stitch — what is formed

The **two-thread chain stitch (ISO stitch type 401)** is formed by a **needle thread** and a **looper thread**. There is no bobbin: the looper thread is supplied from a cone below the machine, so the machine never stops for bobbin changes. The needle thread passes through the fabric as a loop; below the fabric the looper passes its thread through that loop; at the next stitch the needle descends through the looper-thread loop, and so on — each stitch is chained through the previous one. The result: on the top face, a straight line like a lock stitch; on the underside, a row of thread loops — the chain that gives the stitch its name.

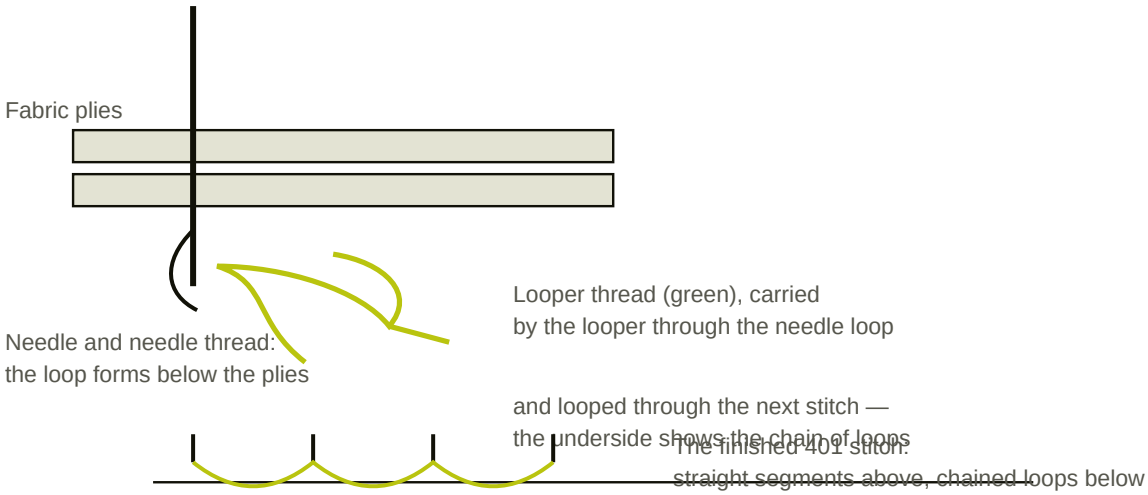


Figure 1 — Chain-stitch formation (schematic cross-section) and the finished stitch: the looper thread interloops with each needle-thread loop below the fabric, chaining the stitches together.

3. Formation of the stitch, step by step

1. **Penetration and loop.** The needle descends through the plies and rises slightly, forming a loop of needle thread below the fabric, as in the lock stitch.
2. **Looper entry.** The looper — a curved, thread-carrying finger below the needle plate that swings in an elliptical path — enters the needle loop, carrying the looper thread through it.
3. **Needle through the looper loop.** The fabric feeds one stitch length; the looper holds its thread out as a triangle of thread, and the descending needle passes inside that triangle, through the looper-thread loop.
4. **Interlooping.** The looper withdraws, the previous needle loop is drawn up around the looper thread, and the new needle loop is taken by the looper in its turn. Each needle loop is thus locked by the looper thread, and each looper loop by the next needle loop — the chain.
5. **Tightening.** Thread pull-up mechanisms set the needle-thread tension so the top face lies flat; the underside loops remain visibly chained, which is normal and correct for 401.

> **THE ONE-WAY WARNING**
>
> A chain stitch is secure only from the finishing end. Pull the looper thread from the *starting* end of a 401 seam and the whole seam runs out stitch by stitch — the property used deliberately when basting, and guarded against in production by back-tacking or locking the seam ends and by sewing direction planning (Study 10).

4. Producing chain-stitch samples

1. Thread the machine: needle thread as for the lockstitch machine (Study 6), and the looper thread through its guides and tension to the looper eye — a missed looper guide is the commonest cause of a machine that "will not chain".
2. Sew samples at the set stitch length on two plies; examine both faces: straight segments above, even chained loops below, no skipped interloops.
3. Measure SPI exactly as in Study 7 (count over a measured length; e.g. 30 stitches in 7.6 cm = 3 in gives SPI = 30 ÷ 3 = 10).
4. Demonstrate the seam's behaviour: unpick a sample from the starting end by pulling the looper thread, and note how the seam ends of production work are protected (back-tack, or the seam crossed by a later operation).

5. Lock stitch (301) versus chain stitch (401)

FEATURE	LOCK STITCH 301	CHAIN STITCH 401
Threads	Two: needle thread + bobbin thread	Two: needle thread + looper thread
Lower mechanism	Rotary hook and bobbin — bobbin holds a limited length and must be changed/refilled	Looper fed from a cone — no bobbin stops, suited to long seams
Interlacing	Threads interlace inside the fabric plies	Threads interloop below the fabric
Appearance	Same straight line on both faces	Straight line on top; chain of loops underneath
Security	Each stitch independent — a broken stitch does not run	Runs out from the starting end if the looper thread is pulled and ends are unsecured
Elasticity	Low — the seam extends little	Higher — the chained loops allow the seam to stretch more
Thread consumption	Lower (roughly 2.5× the seam length in total thread)	Higher (roughly 4× the seam length) — the underside chain costs thread
Typical uses	General seaming of shirts and garments, topstitching, any seam seen from both sides	Long production seams: trouser inseams and side seams, shirt side seams and armhole closing on feed-off-the-arm machines, waistband attaching, felling

6. Common faults in chain-stitch samples

FAULT	LIKELY CAUSES
Stitch will not form / no chain	Looper thread not through its eye or a guide missed; looper timing disturbed; needle fitted wrongly so the looper misses its loop
Skipped stitches	Bent or blunt needle, loop size wrong from a tension/thread mismatch, fabric flagging (lifting with the needle) on light material
Loose, uneven underside loops	Looper-thread tension too low or needle tension too high; uneven feeding
Seam runs out in handling	Ends not secured — a 401 sample without back-tack or end locking behaves exactly as designed; the fault is the missing protection, not the stitch

7. Precautions

- Never pull a chain-stitched sample from the starting end across the record table "to test it" — the demonstration is done once, on a scrap sample made for the purpose.
- Secure the ends of every sample seam as production seams are secured, so the recorded samples survive handling.
- Apply the same needle discipline as Study 6 and 7: new needle first when faults appear.
- Check the looper for burrs with a fingernail whenever the looper thread frays — the looper touches the thread on every stitch.

8. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State the ISO type number of the two-thread chain stitch and the two threads that form it.
2. What replaces the bobbin in a chain stitch machine, and what production advantage follows?
3. Describe the appearance of a correct 401 seam on its two faces.
4. From which end does an unsecured 401 seam run out, and how is this prevented in production?
5. A sample counts 30 stitches in 7.6 cm. Calculate its SPI.
6. Give two seams of a shirt or trouser where 401 is preferred and state the reason in each case.
7. Why does the chain stitch consume more thread than the lock stitch for the same seam length?

Answer key

1. Type 401; the needle thread and the looper thread, interlooped below the fabric.

2. A looper supplied from a cone. The machine never stops for bobbin changes and never runs out of lower thread mid-seam, suiting it to long, continuous production seams.
3. Top face: a straight line of thread segments, like a lock stitch. Underside: a continuous chain of loops formed by the looper thread passing through each needle loop.
4. From the starting end, by pulling the looper thread. Prevented by back-tacking, by locking stitches at the ends, or by arranging for a later seam to cross and secure the ends.
5. 7.6 cm ≈ 3.0 in (7.6 ÷ 2.54 = 2.99). SPI = 30 ÷ 3.0 = 10 stitches per inch.
6. Examples: a long side seam or inseam (no bobbin stops, extra seam elasticity for a stress seam); felling or waistband attaching on a feed-off-the-arm/cylinder machine built around the looper principle. Any two with the no-bobbin or elasticity reason are acceptable.
7. Because every stitch carries, in addition to the needle thread, a loop of looper thread travelling along the underside and around each needle loop; the chained path is longer per stitch than the lock stitch's single interlacing inside the plies.

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